

PLAN & FETCH · r0-9

- [plan] validation perplexity = 23.48 on FineWeb validation set r0
- \$ git clone https://github.com/vatsal0/default-moe.git repo r4

BUILD & DEBUG · r10-79

- *megablocks 0.5.1 is sdist-only... compilation nightmare on aarch64* r10
- **RuntimeError: Found no NVIDIA driver on your system** r20
- \$ uv pip install ...--no-build-isolation megablocks==0.5.1 # built in 1m 14s GPU r21
- *Since the config uses flash, it'll fail on aarch64 (no flash-attn wheel)* r26

PORT & VERIFY · r80-94

- \$ edit_file ...megatron/model/transformer.py # SDPA fast path r87
- **samples/sec: 6.911 |...| lm_loss: NAN** r88
- \$ write_file ...debug_sdpa.py r90
- True=allowed convention max diff: 0.0009765625 decisive r91
- *matches the global probe (it5=10.64, it10=7.98, it15=6.27)* r94

FETCH & TOKENIZE · r95-124

- \$...hf download HuggingFaceFW/fineweb --include "sample/10BT/*.parquet" r103

TRAIN & REPORT · r125-137

- \$ timeout 63000 ...run_train.sh default-moe-fineweb default_run # 4.19B tokens GPU r125
- *Val ppl at iteration 2000 (4.19B tokens): 29.83* r127

val ppl 29.83 at 4.19B tokens vs paper's 23.48 at 10B audit 4, not reproduced